

An Auditable Market-Microstructure Framework for Spoofing-Compatible Sequence Detection

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Abstract

How can surveillance data isolate order-book sequences that are compatible with spoofing without treating ordinary liquidity provision as manipulation? We address this question with an auditable microstructure framework that reconstructs the visible limit order book and groups related execution messages into a single analytical execution. An execution is anchored either by the actor’s passive resting-order execution or by the actor’s aggressive execution. We attribute orders through the original client code when it is available and otherwise through the firm identifier, while preserving the identity level. For each execution, the candidate set comprises the same actor’s opposite-side orders that are active in the top 10 ranks immediately before the first execution and were first observed no more than 90 seconds earlier. The strict gate passes only when four conditions hold simultaneously: cancellation of at least one candidate order is canonically assigned within 2 seconds; attributed withdrawal exceeds execution quantity; observed favorable pre-execution midprice movement is positive; and observed cancel-anchored midprice reversion is positive over a 2-second horizon. An empirical depth kernel weights visible liquidity by book rank. In the Risanamento, Nexi, and Ferrari samples, respectively, the reconstruction identifies 205, 1,684, and 2,475 matched executions, of which 12, 231, and 370 satisfy the strict gate. Both passive and aggressive execution routes contribute to these sets. An exploratory comparison with 36 timed external alert records yields 34 distinct subject–instrument intervals after overlapping or endpoint-sharing records are merged. Every interval contains a recovered execution, 30 contain an attributed withdrawal, and four contain a strict sequence. The four strict overlaps occur only at firm scope, where the external firm maps to multiple canonical actors. These findings support reproducible surveillance triage, but they do not constitute a statistical test, identify manipulative intent, estimate causal effects, measure spoofing prevalence, or validate recall against independently adjudicated manipulation labels. Legitimate liquidity provision remains an explicit alternative explanation.

Keywords. Spoofing detection; market manipulation; limit order book; high-frequency trading; anomaly detection; market maker; market microstructure

1 Introduction

Electronic continuous double auctions organize price discovery through a limit order book (LOB). The LOB records the visible bid and ask orders submitted by market participants and changes whenever an order is entered, amended, executed, or cancelled. Algorithmic and high-frequency trading have increased the speed of these updates and can improve short-term liquidity. The same speed permits large displayed positions to be submitted and withdrawn over intervals that can be difficult to assess as complete sequences in real time.

Spoofing exploits this sequencing problem. In a stylized account, a participant submits a large order on one side of the LOB that is not intended to execute. The displayed quantity may change observed imbalance or microprice indicators and may prompt other participants to revise quotes or submit aggressive orders. The participant can then execute a smaller order on the opposite side at a more favorable price and withdraw the large displayed order. The Market Abuse Regulation (MAR) prohibits orders and behavior that create false or misleading signals about the supply of, demand for, or price of a financial instrument [3].

The observable actions in this mechanism are not unique to spoofing. Liquidity providers also submit, amend, and cancel orders at high frequency, and they may have high order-to-trade ratios. They remove quotes to update prices, control inventory, or limit adverse selection. Static size thresholds, cancellation ratios, and volumetric triggers therefore cannot by themselves distinguish manipulative conduct from legitimate liquidity provision. Such rules can produce false positives and cannot establish the participant’s intent.

A deliberately broad diagnostic makes this overlap visible. [Figure 1](#) ranks actors by the rate at which they cancel liquidity on the side opposite to an execution within the next two seconds. Positive rates occur in all three samples, but the frequency and the number of supporting executions vary substantially across actors. The statistic is deliberately agnostic about why an order was cancelled. A high value can arise from ordinary quote and inventory management, so it is not evidence of manipulation. The dispersion instead motivates a reconstruction that asks which rapid cancellations also belong to the more specific sequence implied by the hypothesized mechanism.

This paper asks whether the observable conjunction implied by the hypothesized mechanism can be reconstructed from participant-level messages without treating frequent cancellation itself as evidence of spoofing. Complete top-10 resting-profile measures first describe the shape of an actor’s displayed liquidity; these diagnostics do not determine eligibility. For each passive or aggressive execution, we then fix a candidate set of same-actor, opposite-side orders that are active in the top 10 ranks and no more than 90 seconds old. A matched execution has at least one cancellation of a candidate order canonically assigned within 2 seconds. The strict four-condition gate additionally requires the attributed withdrawal to exceed execution quantity, observed favorable pre-execution midprice movement to be positive, and observed cancel-anchored midprice reversion to be positive over the 2-second reversion horizon. Actor identity is a separate data problem, so we use the original client code when present and the firm identifier only as a fallback. Each physical cancellation is assigned to at most one execution.

The empirical reconstruction identifies rapid same-actor withdrawals in all three samples and after both execution routes. The full four-condition sequence is less common because it

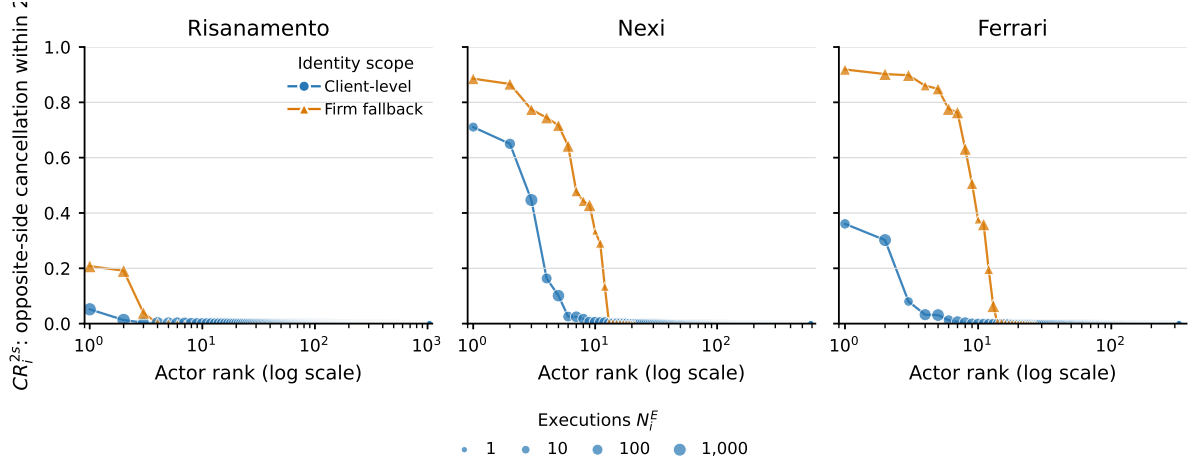


Figure 1: Rate at which each actor cancels opposite-side liquidity within two seconds of its own execution. For actor i , the denominator N_i^E contains all of its passive and aggressive executions. The numerator counts executions followed by at least one cancellation by the same canonical actor on the opposite side during $(t_c^{\text{end}}, t_c^{\text{end}} + 2 \text{ s}]$. The cancellation must also follow the final constituent execution message in canonical source order and remain in the same trading partition. Each execution contributes at most one unit to the numerator, regardless of how many qualifying cancellations follow it. Each execution defines its own window, so the same cancellation can follow more than one nearby execution. Client-level and firm-fallback actors form two separate curves and are ranked from highest to lowest rate within each identity scope and instrument. The actor-rank axis is logarithmic, and no actor identifier is displayed. Identity follows the client-first rule: the original client code is used when available, with the firm identifier used only as a labelled fallback. Marker area increases with $\log(1 + N_i^E)$ to make sample support visible; marker shape and color also distinguish the two identity scopes. No actor is removed for low activity. Firm fallback is a coarser scope that can pool client activity, so its rates are not directly equivalent to client-level rates. This broad statistic does not impose the later conditions on candidate age, the fixed candidate set, withdrawal size, or the price path, and it does not use WMSCI, MSCI, or the strict gate. A high rate is not evidence of manipulation.

additionally requires attributed withdrawal to exceed execution quantity, favorable movement, and reversion. This distinction is the main interpretive result: a mechanical withdrawal can motivate review, but it is not equivalent to a sequence that also satisfies the size and price-path conditions of the stylized mechanism. The output is therefore an auditable set of descriptive screening diagnostics for analyst review, not a statistical validation, a label of manipulation, or an estimate of causal price impact. Within the candidate set, the Withdrawal Multilevel Spoofing Conditionality Index (WMSCI) is the primary event-prioritization metric. It ranks events by the scale and speed of attributed withdrawal relative to the execution, but it does not define the four-condition gate.

An exploratory comparison with externally supplied surveillance alerts gives the same evidence hierarchy. The 36 timed source records form 34 distinct subject–instrument intervals after overlapping or endpoint-sharing records are merged. All 34 intervals contain a recovered execution, 30 contain an attributed withdrawal, and four contain a strict sequence. All four strict overlaps occur where the external firm-level subject maps to multiple canonical actors, so they establish subject-scope coverage rather than exact-actor correspondence. Because the external alerts are positive seeds rather than independently adjudicated labels, this comparison reports recovery relative to the source alerts. Exact-actor strict recall is defined only for intervals whose external and internal identities have aligned granularity; it is not recall against confirmed manipulation.

2 Market Making Versus Spoofing

The screening model starts from a contrast between two stylized economic benchmarks. Both can generate many order messages and cancellations, but they assign different hypothesized roles to displayed liquidity. This contrast does not imply that every market maker quotes symmetrically or that every spoofing episode follows one fixed pattern. It identifies observable features that can guide surveillance.

2.1 The Market Maker’s Problem

A market maker supplies liquidity by posting resting orders on the bid and ask sides of the book. In a stylized inventory-risk benchmark, the dealer seeks to earn the bid–ask spread while controlling inventory risk and exposure to adverse selection. Optimal quotes therefore depend on both order-arrival rates and the dealer’s inventory.

Stochastic-control models often express this problem through a Hamilton–Jacobi–Bellman equation. The dealer chooses bid and ask quotes that balance execution revenue against inventory risk. For example, a long inventory exposes the dealer to a price decline. The dealer can respond by shifting both quotes downward: a lower ask encourages purchases from the dealer, while a lower bid discourages further sales to the dealer. The quote adjustment depends on the midprice, the spread required for compensation, and the penalty for inventory away from the target.

Within this benchmark, frequent cancellation is a risk-management action. The market maker cancels and reposts orders as the midprice changes, as adverse-selection risk changes,

or as inventory moves away from its target. The dealer generally maintains a two-sided presence near the top of the book within this stylized benchmark, although actual quoting can become asymmetric because of inventory, information, or market conditions. Expected profits in the benchmark come primarily from spread capture rather than from deliberately inducing a directional change in the midprice.

2.2 The Spoofer’s Problem

In the stylized spoofing mechanism, the participant is hypothesized to seek to improve the price of a directional execution. The participant displays a large order on one side of the book and submits a smaller order on the opposite side. The large displayed order creates an asymmetric depth profile. Algorithms or traders that use bid–ask volume imbalance may interpret this profile as evidence of short-term price pressure and adjust their orders. If other traders respond to the displayed asymmetry, the resulting price movement could improve the execution of the smaller opposite-side order.

The displayed order creates a trade-off between influence and execution risk. An order near the best quote is prominent in the book but is also more likely to execute. An order posted at Level 2, Level 3, or deeper has a buffer against immediate execution while remaining visible in multilevel order-book data. A participant implementing the stylized strategy may therefore distribute the large profile across deeper levels rather than place all of it at the touch.

After the smaller order executes, rapid withdrawal of the displayed opposite-side profile is consistent with reducing subsequent execution exposure under the hypothesized strategy. This timing motivates the diagnostic contrast with ordinary quote updates: the candidate withdrawal follows the opposite-side execution rather than only changes in fair value, inventory, or adverse-selection risk. [Table 1](#) summarizes these stylized incentives; none is directly observed as intent.

3 Existing Approaches

Spoofing-detection methods differ mainly in the object they classify. Rule-based methods reconstruct a stylized episode from order-book mechanics. Statistical methods monitor unusual states of order flow. Sequence and graph models learn temporal or relational patterns from event data. For the present setting, this taxonomy is useful because it distinguishes methods that reconstruct attributable order sequences from methods that detect unusual states or learned temporal patterns. [Figure 2](#) summarizes the method families, which can overlap in practice.

Early empirical studies operationalize spoofing-surveillance screens through interpretable order-book conditions. Using account-level Korea Exchange data, [\[7\]](#) identify large orders placed far from the best quote, followed by opposite-side trading and subsequent withdrawal. Related surveillance rules have been applied to Brazilian equities [\[9\]](#), FX spot markets [\[11\]](#), cross-market FX activity [\[10\]](#), and Taiwan derivatives [\[1\]](#). These studies connect a screen to observable features such as order size, posting distance, cancellation timing, opposite-side execution, and short-lived price impact. However, the reported implementations use venue-specific thresholds, and their reconstruction can require privileged message data or account identifiers.

A second line of work monitors statistical anomalies in the order book. [\[13\]](#) derive spoofing

Table 1: Stylized differences used to construct the surveillance diagnostics. These differences describe economic benchmarks, not observed intent or necessary conditions for either activity.

Microstructure Feature	Legitimate Market Maker	Hypothesized Spoofing-Compatible Mechanism
Primary objective	Earn the bid–ask spread while controlling inventory and adverse-selection risk.	Improve a directional execution through a displayed order-book signal.
Order-book imbalance	Usually maintains two-sided liquidity, with asymmetry arising from inventory and risk management.	A large, short-lived asymmetry forms the candidate posture.
Cancellation motive	Update quotes to track fair value, control inventory, or limit adverse selection.	Withdrawal after the smaller opposite-side execution is consistent with the mechanism.
Posting-distance trade-off	Quote near the touch to increase execution probability and earn the spread.	Remain visible while reducing the risk that the displayed posture executes.
Source of expected gain	Spread capture, without intentionally moving the midprice in a chosen direction.	A potential favorable price change for the smaller order on the opposite side.

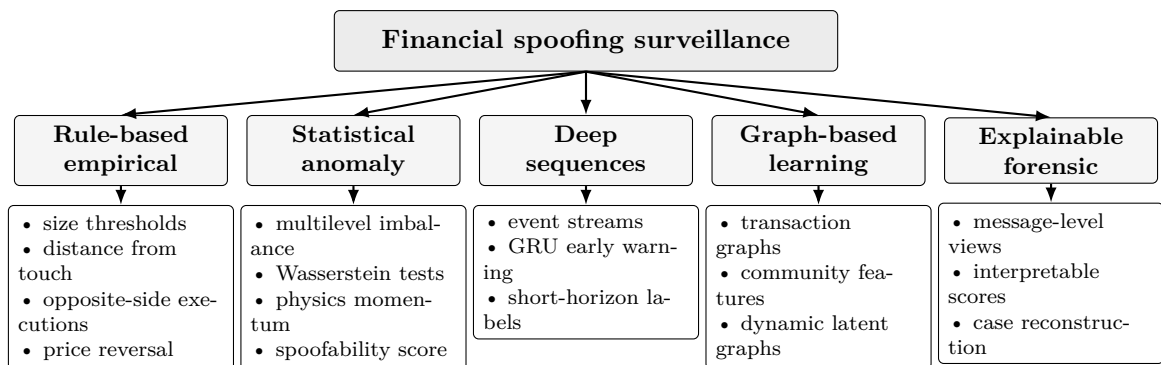


Figure 2: Method families in the spoofing-detection literature. The categories can overlap; each column identifies the main object or representation used by a strand of work.

incentives from multilevel imbalance and propose a Wasserstein-distance statistic for real-time monitoring. [8] represent the order book as a statistical-physics system and detect abnormal momentum in cryptocurrency markets. Explainable forensic methods partly overlap with this family. Message-level visualization can reconstruct prosecuted episodes [2]. Interpretable probabilistic neural networks instead estimate how an isolated limit order changes the conditional distribution of future prices and the expected gain from a hypothetical manipulation strategy in centralized cryptocurrency markets [4]. The latter object measures order-level *spoofability*; it does not establish whether one participant generated the candidate posture, the smaller execution-side order, and the withdrawal. Our framework addresses this complementary attribution problem by linking these observed actions through canonical actor keys while recording whether each linkage is at client or firm scope.

Machine-learning methods add temporal or relational structure. [14] formulate spoofing as an early-warning problem and train a gated recurrent unit model on cryptocurrency order-book event streams. Graph-based methods move the unit of analysis from isolated orders to networks of traders and transactions. This representation can capture coordinated spoofing patterns [5, 6, 15]. Reported models combine temporal transaction graphs, community features, and dynamic latent representations. Their practical performance remains difficult to compare because available labels, market settings, and validation designs differ.

These studies imply two requirements for the present surveillance setting. First, the features must retain their microstructure interpretation. The screen must record candidate volume, posting distance from the touch, cancellation timing, and the relation to an opposite-side execution. Second, any eventual validation must reflect scarce and noisy enforcement labels, rare positive cases, and the analyst cost of false positives. Given the evidence currently available, the model below preserves the auditability of rule-based surveillance while producing descriptive participant-level order sequences from proprietary event data.

4 Data

We use three CONSOB limit-order-book message samples. The Risanamento sample covers June–November 2024, the Nexi sample covers 23 April 2024, and the Ferrari sample covers 13 June 2024. These samples differ in duration and participant composition, so cross-instrument differences are descriptive rather than controlled comparisons.

For an exploratory coverage audit, we also use an externally supplied CONSOB surveillance report containing 37 alert records supplied as positive seeds. These externally identified periods are used to test whether the reconstruction covers the reported activity; they are not independently adjudicated manipulation labels. The source contains 35 timed Ferrari records, one timed Nexi record, and one Risanamento record that identifies only the trading date. We preserve every source record and merge timed intervals that overlap or share an endpoint only within the same instrument and external subject. This procedure yields 34 distinct timed intervals, while the date-only Risanamento record remains outside the event-level denominator.

For timed records, we parse the source bounds at millisecond precision and include both endpoints. To place raw events on a common event-time axis, we use the first nonmissing timestamp in the source-field priority order `TRADETIME`, `BOOKOUTTIME`, `BOOKIN`, and `SEQUENCETIME`.

We recover an execution when at least one of its constituent execution messages lies inside the closed union interval; matched-withdrawal and strict status are then read from that execution’s event-level diagnostics. The union interval, rather than each potentially overlapping source record, is therefore the observational unit for timed counts.

We match each external subject in the identity namespace reported by the source. Consequently, an external firm can cover several canonical actor keys when client identifiers are available internally. We therefore distinguish subject-scope overlap from exact-actor correspondence and treat the latter as identifiable only when the external identity maps to one canonical actor at the same granularity. The alert report contains no negative sample, so the comparison cannot estimate a false-positive rate.

For each instrument, we reconstruct the visible order book from order-level events and infer the tick size from the reconstructed best quotes. We compute actor-wide resting-profile diagnostics over the first ten book ranks and separately construct an age-conditioned set of opposite-side candidate orders. The main specification uses a 10-second structural post-execution window, a 2-second rapid-withdrawal attribution window, a 2-second reversion horizon measured from each assigned cancellation, and a maximum candidate-order age of 90 seconds.

The original client code is not available on every message. We therefore define one canonical actor key before constructing events. If the original client code is present, it defines the actor; only when it is missing do we use the firm identifier. The technical client value 0 is treated as missing rather than as an actor, so a message carrying this sentinel uses the firm fallback when a firm identifier is available. The two namespaces are kept distinct, and every output records whether attribution is at the client or firm level. A client-derived key and a firm-fallback key are distinct canonical identities; the fallback does not merge a client-identified message with a firm-only message. Messages missing both identifiers are excluded from actor-attributed analysis; none occurs in the reported samples. Firm fallback is coarser than client attribution and can combine several clients represented by the same firm. It is therefore an identity fallback, not evidence that those clients are one economic account.

Execution role is defined independently of actor identity. We retain individual execution messages marked exclusively passive or exclusively aggressive and reject messages for which the two role flags are both set or both absent. Related execution messages are grouped into one analytical execution when they share the trading partition—trading date, venue, market, instrument, and EMM code—as well as canonical actor, order identifier, execution role, and side. Passive executions also require a common resting price; aggressive executions can span several traded prices under the same order and sweep. Consecutive constituent execution messages must be no more than 100 milliseconds apart, with no intervening cancellation, modification, or new-order lifecycle event for the same order. The 100-millisecond threshold is an operational choice, and the reported findings are conditional on it; sensitivity to alternative inter-execution-message gaps is not reported here. Raw-message identifiers and the constituent messages assigned to each execution remain available as provenance, while executions form the analytical unit for event metrics and actor-partition aggregation. The resulting surveillance dossiers support analyst review and are not enforcement labels.

5 Model

The model translates the stylized contrast between market making and spoofing into a sequence of observable conditions. Its central question is whether recent opposite-side orders disappear specifically after the same participant executes. To answer it, we construct two distinct objects. The first is the participant’s complete resting profile across the top n book levels, which supplies descriptive DWI, SCI, and MSCI shape diagnostics. The second is an age-conditioned set of opposite-side candidate orders, which supplies matched-withdrawal quantities and the event-prioritization metric WMSCI. Separately, the strict gate applies the prespecified withdrawal-size, favorable-movement, and reversion conditions. These sign and size restrictions are screening conditions, not estimates of causal influence or intent. The resting-profile, withdrawal, and price-response quantities remain separate because they describe different parts of the mechanism.

The motivating rate in [figure 1](#) stops at the broad temporal association between an execution and a later opposite-side cancellation. The reconstruction below narrows this population in stages. It first requires an attributable recent candidate withdrawal, then compares the withdrawn and executed quantities, and finally checks the favorable-movement and reversion conditions. The broad rate does not enter these later gates.

5.1 A Spatio-Temporal Framework for Spoofing Detection

[Figure 3](#) sketches the hypothesized sequence. The detector later translates each state into observable order, execution, and cancellation records rather than inferring the participant’s purpose.



Figure 3: Schematic top- n timeline for the hypothesized spoofing mechanism. The large candidate posture forms an agent-specific depth profile over several bid levels, while the smaller order rests near the opposite best quote. The construction applies symmetrically to candidate buy- and sell-intent episodes. The figure shows the passive execution route; the operational reconstruction also admits an aggressive execution by the same actor.

The method represents a candidate episode as a conditional change in the participant’s LOB footprint. A single order at the best bid or ask does not capture this change. We instead use the participant’s liquidity profile over the first n price levels. This representation includes postures placed at Level 2, Level 3, or across adjacent levels, where displayed liquidity can remain visible while facing less immediate execution risk.

The hypothesized mechanism for participant i has three states:

1. **State 1 (Candidate Posturing):** Shortly before the execution, the participant displays a large liquidity profile on one side of the book, typically away from the touch. The

actor has an opposite-side execution anchor, which may be passive or aggressive in the operational reconstruction.

2. **State 2 (Execution):** The large profile is present when the smaller opposite-side order executes. In the hypothesized mechanism, the profile can coincide with a more extreme measured imbalance before this execution.
3. **State 3 (Withdrawal):** The participant rapidly cancels the opposite-side profile after the execution; under the stylized mechanism, this timing is consistent with reducing subsequent execution exposure.

State 2 has two observable routes. In the passive route, the actor’s resting order executes. In the aggressive route, the actor initiates the trade against resting liquidity. The role flag changes how execution price and quantity are read and how individual execution messages are grouped; it does not change actor identity. In both routes, the candidate posture is sought among the same actor’s recent orders on the side opposite to the actor’s execution side. An aggressive anchor is therefore not itself evidence of spoofing, just as a passive anchor is not evidence of legitimate market making.

These three states illustrate the hypothesized mechanism; their qualitative descriptors do not define strict-set membership. For either execution route, the candidate set consists of same-actor, opposite-side orders observed in the top 10 ranks before the execution and satisfying the 90-second age limit. A cancellation within 2 seconds supplies rapid matched-withdrawal evidence. The strict gate then requires attributed withdrawal to exceed execution quantity and both midprice response diagnostics to be positive. The detector observes these conditions, not the participant’s intent or the profile’s causal effect on other traders. The recency restriction prevents long-lived liquidity from entering a candidate episode solely because it was already resting in the book.

5.1.1 Top- n Participant-Specific Depth Profiles

We first define the participant-specific depth profile used throughout the model. Let $s \in \{b, a\}$ denote the bid or ask side, and let $k = 1, \dots, n$ denote book rank. Rank $k = 1$ is the best quote on side s , and $V_k^{i,s}(t)$ is participant i ’s displayed volume at time t on that side and rank. The vector

$$\mathbf{V}_n^{i,s}(t) = \left(V_1^{i,s}(t), \dots, V_n^{i,s}(t) \right) \quad (1)$$

collects these quantities over the first n ranks. The vector preserves where liquidity is displayed rather than reducing the participant’s entire side of the book to one total. This distinction matters because a large order near the touch has different execution risk from the same quantity posted at a deeper rank.

Raw displayed volume may not be comparable across assets or intraday periods with different market depth. The relative profile therefore divides participant volume by total displayed market

depth at the same side and rank:

$$\tilde{V}_k^{i,s}(t) = \begin{cases} \frac{V_k^{i,s}(t)}{D_k^s(t)}, & D_k^s(t) > 0, \\ 0, & D_k^s(t) = 0, \end{cases} \quad (2)$$

where $D_k^s(t)$ is the total displayed market depth on side s and rank k . The zero-depth branch makes the rank-level convention explicit; an imbalance remains undefined when the participant has no positive weighted footprint on either side. We denote the distance from the touch by $d_k^s(t)$. In tick units, this distance equals the number of ticks between rank k and the same-side best quote, shifted upward by one. The shift gives Rank 1 a positive distance and preserves the ordering of deeper ranks.

5.1.2 Multilevel Distance-Weighted Imbalance

Economic intuition and a stylized parametric benchmark. We use a depth kernel to reduce the top- n vector to a scalar weighted-liquidity measure. The kernel is designed to assign high weight to ranks that remain represented in multilevel book data while facing comparatively low immediate execution risk. The following parametric kernel illustrates this trade-off:

$$\omega_k^{s,\text{par}}(t) = \frac{e^{-\lambda d_k^s(t)} (1 - e^{-\kappa d_k^s(t)})}{\sum_{\ell=1}^n e^{-\lambda d_\ell^s(t)} (1 - e^{-\kappa d_\ell^s(t)})}, \quad (3)$$

The visibility component $e^{-\lambda d}$ decreases with depth. The protection component $1 - e^{-\kappa d}$ increases as the displayed price moves away from immediate execution. Their product penalizes both the high execution risk near the touch and the lower visibility deeper in the book. The parameters λ and κ control the two shapes. This expression is only a conceptual benchmark: the empirical analysis uses neither these weights nor fitted values of λ and κ . Instead, it constructs rank-level empirical proxies for each instrument and book side.

Figure 4 illustrates the benchmark trade-off and its discrete normalization; it is not an estimated empirical profile.

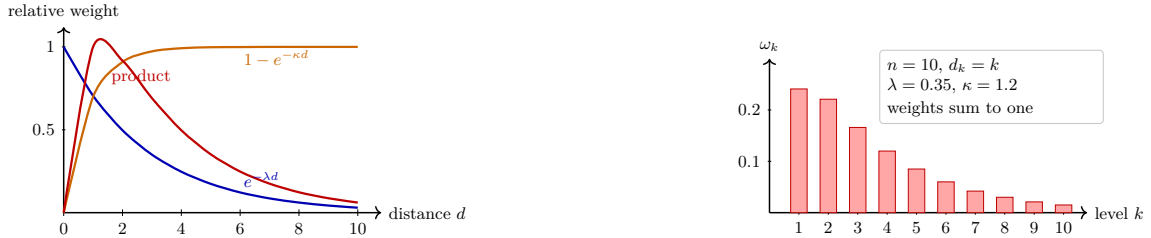


Figure 4: Conceptual depth kernel in equation (3). The left panel shows decreasing visibility, increasing protection from immediate execution, and their product. The right panel shows normalized weights for one illustrative parameter choice. The empirical analysis uses the instrument- and side-specific kernel in equation (10), not these parametric weights.

Operational empirical depth kernel. The operational kernel replaces both exponential components with empirical rank-level quantities. We calibrate each book rank k separately for instrument m and side s . A rank receives more weight when its depth has a strong short-horizon association with midprice changes and its displayed price is rarely reached over the same horizon. Tick distance can vary for a given rank over time, but the calibrated unit remains the rank.

Operational calculation. We reconstruct a post-event top- n snapshot after every ordered message. Let $\mathcal{O}_{m,s,k}$ contain the snapshots in which rank k exists on side s and has positive displayed depth. For each $t \in \mathcal{O}_{m,s,k}$, the indicator $R_{t,k}^{m,s}(H)$ equals one when at least one execution during $(t, t + H]$ occurs at or through the rank's displayed price. An ask rank is reached when the maximum ask execution price in the window is at least the displayed price. A bid rank is reached when the minimum bid execution price is at most the displayed price. We estimate the rank-level message-snapshot hit frequency, denoted $\hat{p}_k^{m,s}(H)$, from pooled hit and exposure counts:

$$\hat{p}_k^{m,s}(H) = \frac{\sum_{t \in \mathcal{O}_{m,s,k}} R_{t,k}^{m,s}(H)}{|\mathcal{O}_{m,s,k}|}. \quad (4)$$

Each observable snapshot contributes once, even when the same rank appears at different tick distances. Thus, [equation \(4\)](#) divides total hits by total exposures; it does not average distance-specific hit rates. Realized tick distance is retained only as descriptive metadata and does not define additional kernel rows or enter a parametric fit. Its probability interpretation is conditional on the message-snapshot sampling scheme. The estimator is a short-horizon proxy for whether the displayed price is reached, not an individual order's execution probability, which also depends on queue position, order size, and intervening cancellations. We define the protection component as the complement of the hit probability:

$$\hat{\rho}_k^{m,s}(H) = 1 - \hat{p}_k^{m,s}(H), \quad (5)$$

A value of $\hat{\rho}_k^{m,s}(H)$ near one indicates that rank k is rarely reached over horizon H .

The second component measures the association between displayed depth and subsequent short-horizon price changes. Let $D_k^{m,s}(t)$ denote total market depth, rather than participant- i depth, at rank k . Total displayed top- n depth at snapshot t is

$$D_n^m(t) = \sum_{r=1}^n \left(D_r^{m,a}(t) + D_r^{m,b}(t) \right). \quad (6)$$

We then define the signed and normalized contribution of side s and rank k :

$$Z_{t,k}^{m,s} = \sigma_s \frac{D_k^{m,s}(t)}{D_n^m(t)}, \quad \sigma_s = \begin{cases} +1, & s = ask, \\ -1, & s = bid. \end{cases} \quad (7)$$

The magnitude $|Z_{t,k}^{m,s}|$ is the fraction of visible top- n depth supplied by that side and rank. The sign makes ask and bid depth contribute in opposite directions to imbalance.

For each snapshot, $M_{t,H}^m$ is the last finite reconstructed midprice observed after t and no later than $t + H$. The corresponding price change is $\Delta M_{t,H}^m = M_{t,H}^m - M^m(t)$. Let $\mathcal{V}_{m,s,k}(H)$ contain the rank- k snapshots for which both midprices are finite. For each instrument-side-rank

cell, we compute the following pooled empirical covariance in absolute value, normalized by the number of eligible snapshots. The overbars denote sample means over $\mathcal{V}_{m,s,k}(H)$:

$$\hat{\nu}_k^{m,s}(H) = \left| \frac{1}{|\mathcal{V}_{m,s,k}(H)|} \sum_{t \in \mathcal{V}_{m,s,k}(H)} \left(Z_{t,k}^{m,s} - \bar{Z}_k^{m,s} \right) \left(\Delta M_{t,H}^m - \overline{\Delta M}_{k,H}^{m,s} \right) \right|. \quad (8)$$

The absolute value makes $\hat{\nu}_k^{m,s}(H)$ a measure of the magnitude, rather than the direction, of rank-specific co-movement. We pool observations across realized tick distances before computing the covariance. We do not compute or average separate distance-bucket covariances. This quantity is a weighting proxy, not a causal estimate of price impact. It is also expressed in instrument-specific price units. We therefore use it only to allocate weights across ranks on the same side of one instrument, not to compare covariance magnitudes across instruments.

The empirical kernel follows the same multiplicative decomposition as [equation \(3\)](#), but no exponential curve is fitted. Before normalization, we enforce a zero lower bound on protection and apply a named 10^{-12} numerical floor to the absolute covariance:

$$\hat{\rho}_k^{m,s}(H) = \max\{1 - \hat{p}_k^{m,s}(H), 0\}, \quad \tilde{\nu}_k^{m,s}(H) = \max\{\hat{\nu}_k^{m,s}(H), 10^{-12}\}, \quad (9)$$

We normalize the product of these two quantities within each instrument and side:

$$\hat{\omega}_k^{m,s}(H) = \frac{\tilde{\nu}_k^{m,s}(H) \hat{\rho}_k^{m,s}(H)}{\sum_{\ell=1}^n \tilde{\nu}_\ell^{m,s}(H) \hat{\rho}_\ell^{m,s}(H)}. \quad (10)$$

The denominator must be finite and strictly positive. If every rank on a side has zero protection, or if the product is otherwise degenerate, calibration fails instead of returning an uninterpretable all-zero vector. The covariance floor is a numerical regularizer rather than empirical evidence of price association. When estimated covariances are at or below the floor, relative weights can be driven primarily by protection and must not be interpreted as evidence of rank-specific price association. Every reported kernel in [table 2](#) passes the normalization condition and sums to one before rounding.

In the remaining formulas, we use the shorthand $\omega_k^s(t) \equiv \hat{\omega}_k^{m,s}(H)$ for an event in instrument m ; the time argument indicates that the fixed weight is applied to a time-varying profile. Every event for instrument m uses the same calibrated kernel. The kernel is not re-estimated by actor, event, or observed outcome, and calibration admits only one weight for each (m, s, k) cell. The descriptive analysis uses the full available sample for each instrument. A prospective or out-of-sample application must instead estimate the kernel on an earlier training period and hold it fixed, so that future order flow does not enter the weights.

The operational calibration uses the first ten ranks and the same $H = 10$ second horizon for every instrument. It produces 20 weights per instrument, one for each side-rank cell. The kernel remains side-specific. To draw one curve per instrument for visualization only, we average the bid and ask weights at each rank:

$$\bar{\omega}_k^m(H) = \frac{1}{2} \left(\hat{\omega}_k^{m,a}(H) + \hat{\omega}_k^{m,b}(H) \right). \quad (11)$$

Table 2: Operational full-sample empirical top-10 depth-kernel weights at horizon $H = 10$ seconds. Values are rounded to three decimal places and sum to one within each instrument and side before rounding. These weights, rather than fitted exponential parameters, enter the empirical spoofing metrics.

Instrument	Side	$\hat{\omega}_1$	$\hat{\omega}_2$	$\hat{\omega}_3$	$\hat{\omega}_4$	$\hat{\omega}_5$	$\hat{\omega}_6$	$\hat{\omega}_7$	$\hat{\omega}_8$	$\hat{\omega}_9$	$\hat{\omega}_{10}$
Risanamento	Ask	0.127	0.248	0.348	0.062	0.004	0.039	0.018	0.012	0.078	0.064
	Bid	0.526	0.012	0.019	0.146	0.091	0.065	0.082	0.034	0.002	0.024
Nexi	Ask	0.028	0.069	0.033	0.138	0.225	0.149	0.275	0.011	0.020	0.051
	Bid	0.045	0.166	0.184	0.276	0.165	0.033	0.006	0.052	0.064	0.008
Ferrari	Ask	0.095	0.022	0.016	0.209	0.270	0.032	0.144	0.032	0.033	0.147
	Bid	0.148	0.107	0.007	0.054	0.072	0.197	0.102	0.038	0.176	0.100

Each side-specific kernel sums to one, so $\sum_{k=1}^{10} \bar{\omega}_k^m(H) = 1$. We use this average only in [figure 5](#). The empirical surveillance metrics use the side-specific top-10 weights in [table 2](#).

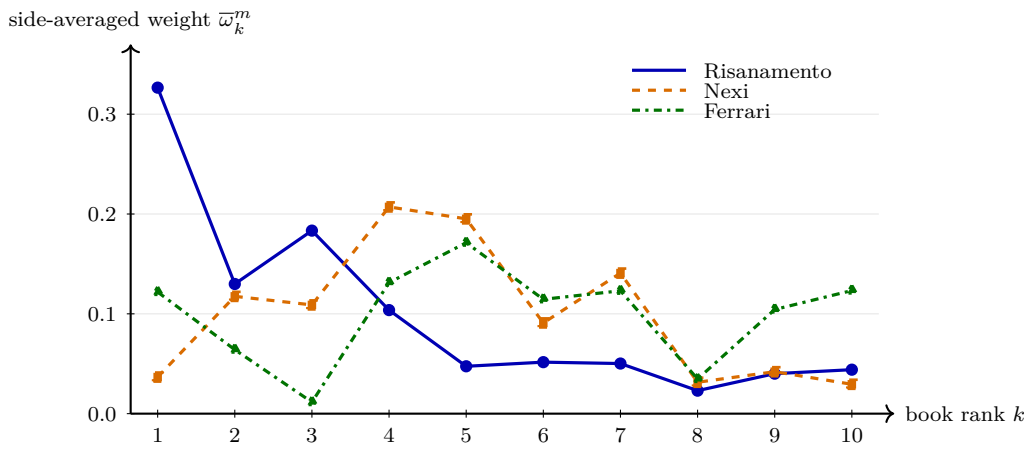


Figure 5: Full-sample empirical top-10 depth-kernel profiles at $H = 10$ seconds. Each line is the arithmetic mean of the separately normalized bid and ask weights in [equation \(11\)](#). The mean permits one curve per instrument on a common rank axis and is used only for visualization. Surveillance metrics use the underlying side-specific kernels.

To expose the two empirical inputs rather than only their normalized product, [Figures 6](#) and [7](#) plot the side-specific visibility and execution-risk profiles used in the calibration. The second plot reports $\hat{p}_k^{m,s}(H)$ directly; the kernel uses its protection complement $\hat{\rho}_k^{m,s}(H) = 1 - \hat{p}_k^{m,s}(H)$.

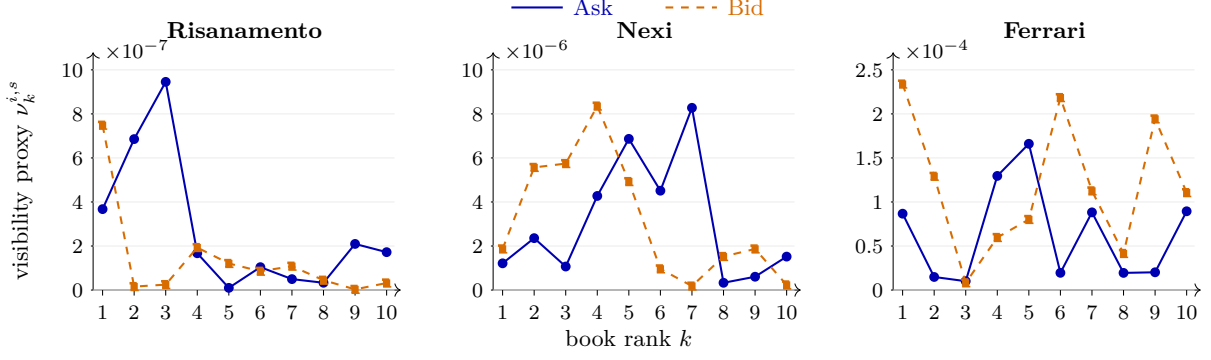


Figure 6: Side-specific empirical visibility proxy by book rank. Each panel shows the stored `visibility_component` values from the full-sample top-10, $h = 10$ -second calibration. Separate axis multipliers preserve the instrument-specific price units; vertical magnitudes should not be compared across instruments.

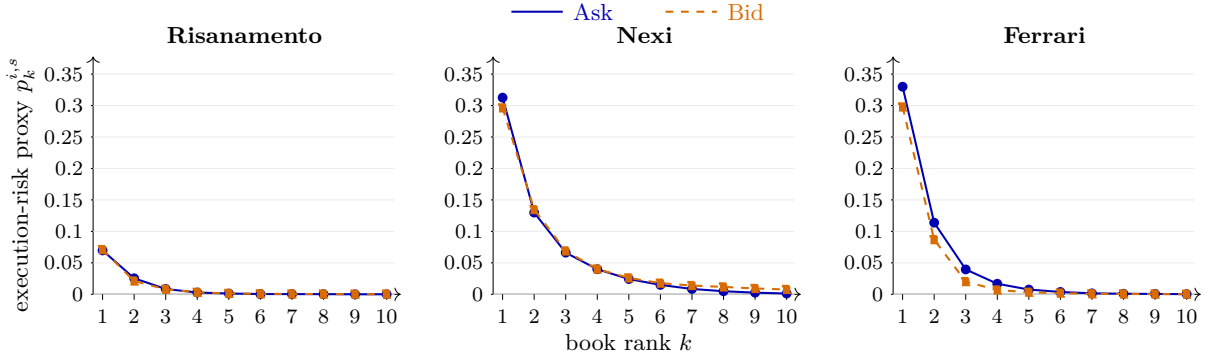


Figure 7: Side-specific empirical execution-risk proxy by book rank. The plotted quantity is the stored h -second reach probability $p_k^{i,s}$ (`hit_probability`) from the full-sample top-10, $h = 10$ -second calibration. The empirical kernel uses its protection complement $\rho_k^{i,s} = 1 - p_k^{i,s}$.

Using the calibrated kernel, we define participant i 's weighted top- n liquidity on side s as

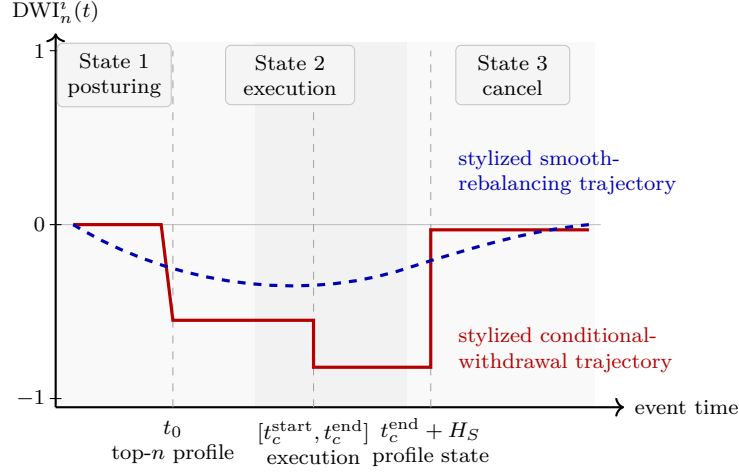
$$L_n^{i,s}(t) = \sum_{k=1}^n \omega_k^s(t) \tilde{V}_k^{i,s}(t). \quad (12)$$

We summarize the difference between the two sides with the *Multilevel Distance-Weighted Imbalance*:

$$\text{DWI}_n^i(t) = \begin{cases} \frac{L_n^{i,a}(t) - L_n^{i,b}(t)}{L_n^{i,a}(t) + L_n^{i,b}(t)}, & L_n^{i,a}(t) + L_n^{i,b}(t) > 0, \\ \text{undefined}, & L_n^{i,a}(t) + L_n^{i,b}(t) = 0. \end{cases} \quad (13)$$

When the denominator is positive, $\text{DWI}_n^i(t) > 0$ denotes an ask-heavy profile and $\text{DWI}_n^i(t) < 0$ denotes a bid-heavy profile. The denominator scales the difference by the participant's total weighted footprint, reducing sensitivity to the absolute scale of that footprint within the same instrument and kernel calibration. Values near $+1$ and -1 correspond to profiles concentrated almost entirely on the ask and bid sides, respectively. A value near zero denotes a balanced positive footprint, whereas the statistic remains undefined when both weighted sides are absent. The reconstructed state series additionally records a one-time zero-profile transition when a

previously observed top- n profile disappears; DWI is set to zero only for that explicit operational state. Setting $n = 1$ gives the best-level measure. Larger values of n also capture candidate liquidity posted behind the touch. DWI is a descriptive profile-shape diagnostic. Neither its level nor its change defines candidate-set membership or the strict gate. Figure 8 illustrates two stylized paths; it is not an empirical classification rule.



For the opposite spoofing direction, the sign pattern is mirrored.

Figure 8: Stylized execution-time behavior of the multilevel distance-weighted imbalance. The conditional-withdrawal trajectory shows a large top- n imbalance returning rapidly toward zero after the opposite-side execution. The smooth trajectory illustrates gradual rebalancing. Neither trajectory is an empirical classification rule.

5.1.3 Conditional Withdrawal Reconstruction

A static multilevel imbalance cannot establish manipulative intent. A market maker can also become temporarily imbalanced while controlling inventory or responding to adverse selection. The additional observable feature is whether the profile disappears conditionally after the participant's execution on the opposite side, as predicted by the stylized mechanism.

Let c denote an execution for canonical actor i . Its ordered constituent execution messages share the trading partition, actor key, order identifier, execution role, and side, and they satisfy the operational inter-execution-message-gap rule. Passive executions also share the resting price. Aggressive executions use reported traded price and quantity and may span several price levels within the same order or sweep. The timestamps t_c^{start} and t_c^{end} denote the first and last constituent execution messages. The quantity Q_c is their total executed quantity, and $s_c \in \{b, a\}$ is the actor's execution side. We denote the opposite candidate-posture side by $\bar{s}_c$. The pre- and post-execution observation times are defined from the reconstructed state series in source order. Each state is recorded after applying its message. Therefore, t_c^- denotes the last actor-state row whose source index is strictly smaller than the first constituent execution message's index. Let $t_c^{\text{target}} = t_c^{\text{end}} + H_S$, where $H_S = 10$ seconds is the structural state horizon. The post-execution state t_c^+ is the last reconstructed state at or before t_c^{target} whose source index is no smaller than the final constituent execution message's index. SCI, each side-specific collapse measure $C_{c,n}^{i,s}$, and MSCI are undefined when either required reconstructed state is unavailable.

We also impose a pre-execution episode window $\Theta > 0$. An opposite-side order enters the candidate profile for execution c only when its first observed posting time τ_o satisfies

$$0 \leq t_c^{\text{start}} - \tau_o \leq \Theta. \quad (14)$$

The empirical analysis uses $\Theta = 90$ seconds as an exploratory default. Let $\mathcal{P}_{c,n,\Theta}^{i,\bar{s}_c}$ contain the orders that are active immediately before the first constituent execution message, belong to actor i , lie on side $\bar{s}_c$ within the first n book ranks, and satisfy [equation \(14\)](#). The detector fixes this candidate set at the first constituent execution message. It records each order's visible quantity, rank, kernel-weighted relative-depth contribution, and first-observed time. In particular, the raw candidate quantity is

$$Q_{c,n}^{i,\bar{s}_c} = \sum_{o \in \mathcal{P}_{c,n,\Theta}^{i,\bar{s}_c}} q_o(t_c^-). \quad (15)$$

The age restriction therefore governs candidate-order attribution, matched withdrawal, WMSCI, and the strict event gate. It does not filter the state series used by SCI and MSCI.

The *Spoofing Conditionality Index* instead measures the absolute change in the normalized shape of actor i 's complete top- n resting profile:

$$\text{SCI}_{c,n}^i = \left| \text{DWI}_n^i(t_c^-) - \text{DWI}_n^i(t_c^+) \right|. \quad (16)$$

An absolute imbalance change does not identify which side lost liquidity. We therefore also measure the fraction of weighted liquidity that disappears from each side after the execution using an exact piecewise definition:

$$C_{c,n}^{i,s} = \begin{cases} \frac{[L_n^{i,s}(t_c^-) - L_n^{i,s}(t_c^+)]_+}{L_n^{i,s}(t_c^-)}, & L_n^{i,s}(t_c^-) > 0, \\ 0, & L_n^{i,s}(t_c^-) = 0. \end{cases} \quad s \in \{b, a\}. \quad (17)$$

where $[x]_+ = \max\{x, 0\}$. A large $C_{c,n}^{i,\bar{s}_c}$ indicates that the actor's top- n liquidity decreased on the side opposite to the execution. A two-sided quote refresh after a midprice change can instead generate material collapse on both sides. The boundary values have a direct interpretation: $C_{c,n}^{i,s} = 1$ means that the entire weighted pre-execution profile on side s disappeared, whereas $C_{c,n}^{i,s} = 0$ means that it did not decrease or was absent before execution.

To summarize these diagnostics while preserving evidence in both directions, define the event-level signed contrast

$$\text{MSCI}_{c,n}^i = \frac{1}{2} \text{SCI}_{c,n}^i + C_{c,n}^{i,\bar{s}_c} - C_{c,n}^{i,s_c}. \quad (18)$$

Because DWI lies in $[-1, 1]$, $\text{SCI}/2$ and each collapse term lie in $[0, 1]$. Therefore, MSCI lies in $[-1, 2]$ without an additional clipping step. A positive MSCI means that normalized shape change and opposite-side collapse outweigh same-side collapse. Zero is the neutral contrast, whereas a negative MSCI means that same-side collapse dominates.

The positive part in the definition of C distinguishes liquidity disappearance from liquidity growth. It does not truncate the subsequent cross-side contrast. The three inputs remain

separately inspectable, so a two-sided quote refresh is distinguishable from an asymmetric collapse. MSCI is a secondary event-level diagnostic that enters actor-level feature summaries. It is neither a probability nor matched-withdrawal evidence. It does not enter the four-condition event gate or the reported result tables. Operational outputs call it `MSCI_resting_profile` to emphasize that it uses all active same-actor top- n orders, not only the age-conditioned candidate set. Consequently, long-lived balanced liquidity can dilute the measured shape change. MSCI should therefore be interpreted as context about the actor’s complete resting profile rather than as the collapse of the recent candidate layer.

5.1.4 Canonical Matched Withdrawal

A cancellation can be eligible for several nearby executions. To measure withdrawal without counting that cancellation more than once, let $\mathcal{E}(o)$ contain the eligible same-actor, opposite-side executions that precede cancellation o by at most $\Delta\tau_W$, where $\Delta\tau_W = 2$ seconds in the reported specification. Eligibility is evaluated only when the cancelled order belongs to the execution’s fixed pre-execution candidate set; the timing and source-sequence precedence conditions stated below must also hold. We assign o deterministically to the canonical execution

$$a(o) = \arg \min_{c \in \mathcal{E}(o)} (t_o - t_c^{\text{end}}, t_c^{\text{start}}, \text{id}_c), \quad (19)$$

where the tuple is ordered lexicographically and the stable execution identifier breaks any final tie. Eligibility also requires source-sequence precedence: the cancellation follows the final constituent execution message in both event time and source sequence. For eligible top- n cancellations, define $\mathcal{C}_{c,n}^i = \{o : a(o) = c\}$. A physical cancellation is identified by its trading partition, source index, and candidate order identifier. We preserve every eligible link for audit, but each physical cancellation contributes to at most one execution’s metrics.

An order can grow after the execution but before its cancellation. Such growth was not part of the pre-execution candidate profile and must not increase the withdrawal attributed to that profile. For each assigned order $o \in \mathcal{C}_{c,n}^i$, let $q_{c,o}^{\text{pre}}$ denote its visible quantity immediately before the execution and let q_o^{cancel} denote its visible quantity immediately before cancellation. We define the attributed quantity as

$$\bar{q}_{c,o} = \min\{q_{c,o}^{\text{pre}}, q_o^{\text{cancel}}\}. \quad (20)$$

The raw cancellation-time quantity remains available for audit. The cap ensures that only liquidity already present in the candidate layer contributes to matched withdrawal. The total assigned withdrawal and its ratio to execution quantity are

$$U_{c,n}^i = \sum_{o \in \mathcal{C}_{c,n}^i} \bar{q}_{c,o}, \quad R_{c,n}^i = \frac{U_{c,n}^i}{Q_c}, \quad (21)$$

where $Q_c > 0$ is the execution quantity by construction. Using the age-conditioned candidate

quantity $Q_{c,n}^{i,\bar{s}_c}$ from [equation \(15\)](#), we report the fraction of that profile removed as

$$A_{c,n}^i = \begin{cases} \frac{U_{c,n}^i}{Q_{c,n}^{i,\bar{s}_c}}, & Q_{c,n}^{i,\bar{s}_c} > 0, \\ \text{undefined}, & Q_{c,n}^{i,\bar{s}_c} = 0. \end{cases} \quad (22)$$

The two ratios answer different questions. $R_{c,n}^i$ compares the attributed withdrawal with the executed quantity, whereas $A_{c,n}^i$ compares it with the candidate profile that was available to withdraw. Under the reconstructed order lifecycle and unique physical-cancellation assignment, the cap in [equation \(20\)](#) keeps $A_{c,n}^i$ in $[0, 1]$ whenever its denominator is positive.

For event prioritization, we also summarize the scale and speed of the attributed withdrawal. Define the delay-weighted withdrawal quantity

$$\tilde{U}_{c,n}^i = \sum_{o \in \mathcal{C}_{c,n}^i} \bar{q}_{c,o} \exp\left(-\frac{t_o - t_c^{\text{end}}}{\tau_w}\right), \quad \tau_w = 10 \text{ seconds}. \quad (23)$$

Within the 2-second assignment window, the exponential factor decreases with cancellation delay. The decay scale τ_w prioritizes temporal proximity; it is not an estimate of a causal response rate. The event-level *Withdrawal Multilevel Spoofing Conditionality Index* is

$$\text{WMSCI}_{c,n}^i = \begin{cases} \log\left(1 + \frac{Q_{c,n}^{i,\bar{s}_c}}{Q_c}\right) \log\left(1 + \frac{\tilde{U}_{c,n}^i}{Q_c}\right) A_{c,n}^i, & Q_{c,n}^{i,\bar{s}_c} > 0 \text{ and } \tilde{U}_{c,n}^i > 0, \\ 0, & Q_{c,n}^{i,\bar{s}_c} = 0 \text{ or } \tilde{U}_{c,n}^i = 0. \end{cases} \quad (24)$$

The first logarithmic term measures the candidate posture relative to the execution. The second measures the rapidly withdrawn quantity on the same scale, and $A_{c,n}^i$ measures the share of the candidate posture that is removed. Under the piecewise definition, WMSCI is zero when no positive candidate-layer or delay-weighted attributed-withdrawal contribution is available; this does not imply that the unrestricted resting profile was unchanged. WMSCI is non-negative and has no fixed upper bound. We use it as the primary metric for ordering event-level candidates for human review. It is neither a probability nor a calibrated measure of manipulative intent, and its magnitude should not be interpreted as a proportional increase in suspiciousness. The full descriptive gate instead requires $U_{c,n}^i > Q_c$ and the separate price-path conditions; it does not apply a WMSCI threshold. At the actor-partition level, we summarize repeated behavior through counts and withdrawal rates rather than aggregating WMSCI into an actor ranking.

5.1.5 Price-Response Diagnostics

The matched-withdrawal variables show whether an actor removes same-actor opposite-side liquidity after its own execution. They do not show whether the price path is consistent with the hypothesized mechanism. Such a path first moves in the direction favorable to the execution and then partly reverses after the candidate layer is removed. We measure these two features with separate price-response diagnostics. They are economic consistency checks, not causal estimates of manipulation.

Let $b(t)$ and $a(t)$ denote the best bid and ask, with corresponding displayed quantities

$q_1^b(t)$ and $q_1^a(t)$. When both quotes and positive top-level depth are available, the two market-price proxies are the midprice $M_{\text{mid}}(t) = [a(t) + b(t)]/2$ and the queue microprice $M_{\text{micro}}(t) = [a(t)q_1^b(t) + b(t)q_1^a(t)]/[q_1^b(t) + q_1^a(t)]$ [12]; otherwise the corresponding diagnostic is missing. Let $M(t)$ denote either proxy, and let t_0 be the earliest first-observed posting time among the eligible candidate orders for execution c . We encode the execution direction as

$$h_c = \begin{cases} +1, & \text{if the execution is a sell execution,} \\ -1, & \text{if the execution is a buy execution.} \end{cases} \quad (25)$$

Under this convention, a positive signed change is favorable to the execution: a sell benefits from an upward move and a buy from a downward move. We measure the favorable movement between candidate-layer posting and execution as

$$\text{FPM}_c^i = h_c [M(t_c^-) - M(t_0)], \quad (26)$$

Reversion is anchored to each assigned cancellation rather than to the end of the execution. Let t_o denote the cancellation timestamp, t_o^- the last reconstructed state strictly before its source index, and $H_R = 2$ seconds the reversion horizon. Let $\mathcal{C}_{c,n}^{i,\text{obs}}$ contain the assigned cancellations for which the relevant pre-cancellation and horizon values of M are observed and $w_{c,o} > 0$ is finite. The execution-level post-cancellation reversion is the quantity-and-delay weighted mean

$$\text{REV}_c^i = \frac{\sum_{o \in \mathcal{C}_{c,n}^{i,\text{obs}}} w_{c,o} h_c [M(t_o^-) - M(t_o + H_R)]}{\sum_{o \in \mathcal{C}_{c,n}^{i,\text{obs}}} w_{c,o}}, \quad w_{c,o} = \bar{q}_{c,o} \exp\left(-\frac{t_o - t_c^{\text{end}}}{\tau_w}\right), \quad (27)$$

with $\tau_w = 10$ seconds in the reported run. Here $M(t_o + H_R)$ denotes the last reconstructed state at or before the target time, provided that the message sequence covers that target. The statistic is undefined when no assigned cancellation contributes a finite positive weight with both required price observations. A positive FPM_c^i means that the price moved in the direction favorable to the execution while the candidate layer was present. A positive REV_c^i means that the market-price proxy moved against the pre-execution favorable direction after the assigned cancellation or cancellations. For quantity-weighted execution price P_c , we also define the local execution advantage relative to the pre-posture benchmark:

$$\text{ADV}_c^i = h_c [P_c - M(t_0)]. \quad (28)$$

The execution advantage is positive when a sell executes above the benchmark or a buy executes below it.

We compute each diagnostic with both the midprice and the queue microprice in the event-level records. The reported summary tables and the event gate use the midprice versions. We report price responses separately from withdrawal counts, quantities, and rates instead of multiplying all components into one detector. Withdrawal variables describe the reconstructed message sequence. The variables FPM, REV, and ADV describe the associated price path. This separation distinguishes repeated conditional withdrawals without a favorable price response from episodes that also exhibit favorable pre-execution movement and post-cancellation

reversion.

5.1.6 Event Reconstruction

We reconstruct events from order-level messages rather than aggregate snapshots. A resting order can change book rank when the best quote moves even though its owner has neither modified nor cancelled it. Snapshot changes alone can therefore confound repricing of the surrounding book with withdrawal by the participant. The reconstruction follows the economic sequence from actor identification to execution, withdrawal, and price response:

1. Resolve the canonical actor and classify each execution message as exclusively passive or aggressive. Construct execution c from compatible constituent execution messages while preserving the ordered mapping to raw messages.
2. Reconstruct all active orders of actor i immediately before t_c^{start} , assign them to side and rank, and compute the unrestricted $L_n^{i,b}(t_c^-)$, $L_n^{i,a}(t_c^-)$, and $\text{DWI}_n^i(t_c^-)$.
3. From the opposite-side top- n orders in this pre-execution state, retain the fixed candidate set $\mathcal{P}_{c,n,\Theta}^{i,\bar{s}_c}$ whose first-observed times satisfy the age restriction.
4. Follow the complete actor state through the structural target $t_c^{\text{end}} + H_S$ and select t_c^+ according to the source-order and event-time rule above.
5. Retain every eligible execution-cancellation link. Assign each physical cancellation canonically using [equation \(19\)](#), exclude competing links from metric totals, and cap attributed quantity at the candidate order’s pre-execution visible quantity using [equation \(20\)](#).
6. Compute SCI, side-specific collapse, and MSCI from the unrestricted resting-profile states at t_c^- and t_c^+ .
7. Report assigned withdrawal quantity, withdrawal-to-execution ratio, removed-profile fraction, and WMSCI. Use WMSCI to order event-level candidates for human review without applying it as a strict-gate threshold.
8. Compute signed price diagnostics from the first candidate-layer posting, through the execution, to the post-cancellation window.
9. Summarize repeated behavior with execution counts and actor-level recurrence diagnostics. Retain raw constituent execution messages as provenance.

This procedure is designed to distinguish ordinary book repricing from withdrawal of participant-attributed candidate liquidity. Canonical assignment ensures that one physical cancellation does not inflate the metrics of several neighboring executions under the stated eligibility rule.

The event-level *spoofing-compatible sequence* gate passes only when four conditions hold simultaneously: (i) at least one eligible opposite-side cancellation is canonically assigned within 2 seconds of the execution; (ii) the finite attributed withdrawal exceeds the positive execution quantity, $U_{c,n}^i > Q_c$; (iii) the observed midprice FPM_c^i is strictly positive; and (iv) the observed cancel-anchored midprice REV_c^i is strictly positive. A missing or right-censored price diagnostic

fails the conjunction. This transparent rule supports analyst triage; it is descriptive and is not a statistical test, manipulation label, or proof of intent.

The reported specification uses a 10-second structural window, 2-second rapid-attribution and cancel-anchored reversion windows, and a 90-second candidate-order age limit. We estimate the empirical top-10 kernel separately for each instrument. Withdrawal and price-response components remain separate rather than being reduced to a scalar composite score.

6 Results

6.1 Cross-Instrument Detection Summary

The broad rate in [figure 1](#) counts any qualifying opposite-side cancellation after an execution. The results below narrow that behavior to cancellations of the fixed, age-conditioned candidate set and assign each physical cancellation canonically. These matched same-actor withdrawals occur in all three samples and after both passive and aggressive executions. The complete strict sequence is necessarily a smaller subset because, in addition to rapid canonical matching, it requires $U_{c,n}^i > Q_c$, positive favorable pre-execution movement, and positive cancel-anchored reversion. [Table 3](#) reports this narrowing under the same actor-wide top-10 state and age-conditioned candidate-order reconstruction. Each instrument uses its full-sample empirical kernel from [table 2](#) with kernel horizon $H = 10$ seconds.

A matched execution is a passive or aggressive execution followed within 2 seconds by a canonically assigned cancellation of a same-actor, opposite-side candidate order. The separate 10-second interval is the structural reconstruction horizon. The attributed cancellation quantity cannot exceed the candidate order’s visible quantity before execution, while the raw cancellation-time quantity remains available for audit. The table distinguishes executions from their raw constituent execution messages and counts each assigned physical cancellation once. The firm-fallback column records the coarser cases in which the original client code is missing. The strict column then retains matched executions that also satisfy the withdrawal-size and two price-response conditions. These quantities are descriptive screening diagnostics, not manipulation labels or evidence of intent.

Table 3: Actor-attributed execution populations and reconstructed withdrawals by execution anchor. Individual execution messages provide provenance; executions are the analytical unit. Firm fallback counts matched executions for which the client code is missing and the firm identifier defines the actor. Assigned cancellations are counted once. Strict is the four-condition spoofing-compatible sequence gate.

Instrument	Anchor	Executions	Execution messages	Matched	Firm fallback	Assigned cancels	Strict
FERRARI	Aggressive	4,333	9,335	1,153	1,131	1,760	165
FERRARI	Passive	8,664	9,329	1,322	1,147	2,009	205
NEXI	Aggressive	3,849	7,425	620	576	944	100
NEXI	Passive	6,663	7,424	1,064	791	1,867	131
RISANAMENTO	Aggressive	9,956	16,029	71	44	84	10
RISANAMENTO	Passive	15,016	15,929	134	114	136	2

The main empirical pattern is a gap between mechanical matching and the complete price-consistent sequence. The reconstruction identifies 2,475 matched executions for Ferrari, 1,684 for

Nexi, and 205 for Risanamento. The strict gate retains 370, 231, and 12, respectively. Favorable pre-execution movement is positive in 54.9%, 52.8%, and 56.6% of observed matched executions, whereas positive reversion occurs in 35.9%, 36.5%, and 21.7%. Thus, the withdrawal condition is not being used as a substitute for the full hypothesized price path. These are marginal shares among observed matched-execution diagnostics and do not decompose strict-set attrition across the jointly required conditions.

These empirical quantities are direct reconstruction counts and conditional descriptive shares. No counterfactual comparison or inferential test is applied. The table therefore does not provide p -values, confidence intervals, or evidence that withdrawal is more likely after the focal actor’s own execution than under an otherwise comparable no-execution condition.

Event-accounting denominators. Table 3 reports raw-execution provenance, execution-anchor composition, firm-fallback counts, and uniquely assigned physical cancellations for each instrument. Raw execution messages provide provenance, whereas executions are the analytical unit for all conditional shares.

6.2 Risanamento

Risanamento has the smallest matched set, and only 12 executions satisfy the strict four-condition gate. Favorable pre-execution movement is nevertheless positive in 56.6% of matched executions with an observed diagnostic. Among matched executions with an observed reversion diagnostic, 21.7% have positive cancel-anchored reversion. The strict gate additionally requires positive favorable pre-execution movement and attributed withdrawal exceeding execution quantity, so these marginal shares do not identify which requirement excludes any particular matched execution.

For each of the 205 matched executions, the review record preserves an execution-first dossier and a reconstructed top-10 event window. These records organize observed facts, counter-evidence, and legitimate alternatives. They are triage aids, not empirical labels or proof of manipulation.

6.3 Nexi

Nexi exhibits matched withdrawals after both execution routes, but the full price-consistent path remains a subset. Reconstruction identifies 1,684 matched executions, of which 231 pass the strict gate. Positive favorable pre-execution movement occurs in 52.8% of matched executions with an observed diagnostic, while positive cancel-anchored reversion occurs in 36.5%.

Differences from the other instruments are not causal contrasts because the samples cover different periods and actor compositions.

6.4 Ferrari

Ferrari has the largest matched set, with contributions from both passive and aggressive execution anchors. Reconstruction identifies 2,475 matched executions, and 370 pass the strict gate. Among observed matched-execution diagnostics, the shares with positive favorable pre-execution movement and positive reversion are 54.9% and 35.9%.

These are descriptive screening outcomes, not manipulation labels or prevalence estimates.

6.5 Actor-Level Attribution and Interpretation

Recurrence is informative only when actor identity and execution role remain explicit. The largest actor–anchor groups differ in identity resolution, matched-execution count, withdrawal scale, and price-path composition. Table 4 therefore reports the three largest groups in each instrument by matched-execution count rather than collapsing them into one actor score. An actor is either an original client code or, only when that code is missing, a firm identifier. W/E maximum is the largest matched withdrawal-to-execution ratio within the group; FPM+ and REV+ are the shares of matched executions with positive midprice favorable movement and reversion.

Table 4: Largest actor–anchor rapid-withdrawal groups by matched-execution count. Client denotes the original client code. Firm fallback denotes aggregation by firm only when that client code is missing. W/E max is the largest assigned withdrawal-to-execution ratio in the group. FPM+ and REV+ are descriptive shares among matched executions with finite midprice diagnostics.

Instrument	Actor	Identity	Anchor	Matched	W/E max	FPM+	REV+
FERRARI	157922_3	Firm fallback	Aggressive	367	60.0	43.6%	32.7%
FERRARI	157922_3	Firm fallback	Passive	257	77.5	47.1%	30.4%
FERRARI	619575_1	Firm fallback	Passive	242	34.5	52.1%	41.7%
NEXI	23523	Client	Passive	248	1200.0	64.5%	25.4%
NEXI	354310	Firm fallback	Passive	233	565.2	59.7%	28.9%
NEXI	628912	Firm fallback	Aggressive	191	1000.0	33.0%	57.1%
RISANAMENTO	628912	Firm fallback	Passive	83	80.0	79.5%	1.2%
RISANAMENTO	625922	Firm fallback	Aggressive	30	26.8	36.7%	56.7%
RISANAMENTO	17295	Client	Aggressive	22	44.4	36.4%	61.9%

The dispersion of FPM+ and REV+ across these groups shows why recurrence and price response must remain separate. A group can contain many matched withdrawals without exhibiting the same reversion pattern as another group. Because firm fallback can combine several clients, a firm-level count has coarser attribution than a client-level count and should not be interpreted as a single-account history.

The primary empirical object is the actor-attributed association between execution and rapid opposite-side withdrawal, with client-level attribution when an original client code is observed and coarser firm-level attribution otherwise. We report its magnitude and repetition descriptively. WMSCI orders event-level candidates for review; MSCI remains a complete-resting-profile context diagnostic; neither metric is a strict-gate threshold or an actor-ranking score. An analyst must also consider the withdrawn order’s execution risk and distance from the touch, bilateral activity, inventory context, price response, and execution advantage. Together, these diagnostics support assessment of consistency with the stylized mechanism while retaining legitimate liquidity provision as an alternative explanation. Neither WMSCI nor a dashboard review category is treated as ground truth.

6.6 Coverage of Externally Supplied Surveillance Alerts

At the externally reported subject scope, the comparison recovers an execution in every distinct timed interval, but the complete behavioral sequence appears in only four intervals. The 36 timed source records become 34 distinct intervals after overlapping or endpoint-sharing records for the same instrument and external subject are merged. All 34 intervals contain raw subject activity and at least one recovered execution, while 30 contain an attributed withdrawal. [Table 5](#) reports the corresponding hierarchy by instrument.

Table 5: Coverage of externally supplied timed alert records. Overlapping or endpoint-sharing records are merged only within the same instrument and external subject. Execution counts intervals containing at least one execution with a constituent execution message inside the closed union interval. Withdrawal and strict-scope columns count intervals containing at least one execution with the corresponding event-level diagnostic. Strict subject scope requires at least one execution in the interval to satisfy all four gate conditions. The final column reports strict overlaps among intervals whose external identity maps to one canonical actor at the same granularity; entries report strict intervals divided by identity-aligned intervals. The date-only Risanamento record is excluded from all timed denominators.

Instrument	Timed source records	Union intervals	Execution	Withdrawal	Strict scope	Strict / aligned intervals
Ferrari	35	33	33	30	4	0/3
Nexi	1	1	1	0	0	0/1
Total	36	34	34	30	4	0/4

All four strict subject-scope overlaps occur within the 30 identity-unaligned Ferrari intervals for which the main externally reported firm maps to two canonical actor keys. Exact-actor correspondence is therefore not identifiable in those intervals. By contrast, the comparison contains four identity-aligned intervals—three for Ferrari and one for Nexi—and none contains a strict sequence. The Nexi interval nevertheless contains one aggressive execution formed from four constituent execution messages, but it has no attributed withdrawal under the stated timing rule.

The Risanamento source identifies only one trading date. On that date, the reported subject has 59 reconstructed executions, including one execution with an attributed withdrawal and no strict sequence. These full-day counts establish that the subject is observable, but they cannot identify correspondence with the specific external event.

These results measure recovery relative to externally reported activity. Exact-actor strict recall is identifiable only for the four identity-aligned unioned timed intervals, of which none contains a strict sequence. The four strict overlaps among the remaining intervals are subject-scope overlaps, not exact-actor recall. Because the source is positive-only and was not independently re-adjudicated for manipulative intent, these quantities do not estimate recall or precision against true manipulation. No negative windows are available to estimate false positives. Moreover, the external alert rule and the strict four-condition gate need not define the same event population. The sparse strict overlap records limited agreement under two operational definitions. It cannot, by itself, identify false negatives, false positives, or the source of disagreement.

7 Discussion and Conclusion

The main empirical finding is that rapid same-actor withdrawal is broader than the complete spoofing-compatible sequence. Matched episodes occur after both passive and aggressive executions in all three samples. Only a subset also exhibits the required withdrawal size, favorable pre-execution movement, and post-cancellation reversion. This distinction matters because it prevents a high cancellation rate or a large withdrawal from standing in for the full mechanism.

The external-alert coverage audit reinforces this separation. The reconstruction covers execution in all 34 distinct timed intervals and attributed withdrawal in 30, whereas only four intervals contain a strict sequence at subject scope. None of the four identity-aligned intervals contains a strict sequence, and the four strict subject-scope overlaps cannot be resolved to one canonical actor. Thus, the strict gate is not a surrogate for the external surveillance rule.

The mathematical construction makes the separation explicit. The quantities $U_{c,n}^i$, $R_{c,n}^i$, and $A_{c,n}^i$ describe how much of the pre-execution candidate layer can be attributed to the withdrawal and how its scale compares with the execution. The quantities FPM_c^i and REV_c^i describe the associated price path. Requiring both types of evidence produces the strict gate, but the conjunction remains a descriptive compatibility rule. It neither estimates spoofing prevalence nor establishes that any retained sequence was manipulative. WMSCI separately prioritizes event-level review by combining candidate-layer scale, rapid attributed withdrawal, and the removed-profile fraction. Unlike MSCI, it is not a complete-resting-profile collapse measure; unlike the strict gate, it is not a pass-fail condition. MSCI is a separate resting-profile diagnostic: it uses all active same-actor top- n orders, so long-lived balanced liquidity can attenuate the measured shape change. It therefore does not establish that the recent candidate layer itself collapsed.

The strongest limitation is the absence of a counterfactual comparison and an independently adjudicated label set. The analysis does not test whether withdrawal is more likely after the focal actor’s execution than after an otherwise comparable non-event, and repeated observations from the same actors or trading periods need not be independent. The external alerts add positive reference periods but no negative controls. They support source-alert recovery and exact-actor strict-recall accounting under the audit’s identity-alignment rule, but they cannot identify recall or precision against independently adjudicated manipulation labels or calibrate detector thresholds. The depth kernels are also calibrated on the same full samples to which they are applied, so their displayed-price associations are descriptive rather than out-of-sample evidence. Right-censored reversion outcomes fail the event gate, and the reported results do not test sensitivity to the 100-millisecond execution grouping threshold, the 90-second candidate age, or the reported 2- and 10-second windows.

Identity and sampling impose further limits. Firm fallback loses within-firm client resolution and can pool several accounts, so firm-level recurrence is not directly comparable with recurrence under a client code. Aggressive execution is an execution-role observation, not evidence about the intent of the surrounding resting orders; ordinary liquidity taking remains an alternative explanation. External firm-level alerts can also span several internal actor keys, which prevents exact-actor evaluation in those intervals. The three instrument samples cover different periods and actor compositions, which rules out causal cross-instrument comparisons.

The contribution is therefore an auditable surveillance workflow rather than an enforcement classifier. It preserves raw message provenance, separates withdrawal from price-response evidence, and exposes the gate conditions used for triage. Prospective use would require a pre-specified counterfactual design and thresholds, kernel calibration on training periods only, chronological out-of-sample validation, and evaluation against independently reviewed cases with explicit false-positive and false-negative costs.

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